

An improved multiple UAVs cooperative flight algorithm based on Leader Follower strategy

Tao Chen, Qi Gao, MingYu Guo

School of Automation, Beijing Institute of Technology, Beijing 100081

E-mail: chet209hds@163.com

Abstract: With the development of micro unmanned aerial vehicles (UAVs), the UAVs have broad prospects and social needs in many fields such as disaster rescue, fire alarm, road reconnaissance, logistics transportation and so on. Compared to the single drone, multiple UAVs cooperation can enhance the efficiency and the capacity on load handling. In this paper, an improved multiple UAVs cooperative flight algorithm is proposed based on Leader Follower strategy. The Ultra Wide Band (UWB) ranging modules are applied to multiple UAVs cooperative flight, the relative distance between the UAVs provided by the UWB modules and the segmented waypoints GPS information are combined to achieve a cooperative flight. The cooperative algorithm consists of two main parts: the relative distance control is based on ESO (extended state observer), which can effectively guarantee the movement consistency and avoid the collision during flight; the transverse correction algorithm can ensure the following UAV track the leader tightly by segmented waypoints GPS information. Compared to the original Leader Follower strategy, this algorithm does not need complex communication protocols and higher real-time information processing ability. Meanwhile, it can achieve cooperative flight even when there exists a larger disturbance.

Key Words: UAV, UWB, cooperative control, ESO

1 Introduction

Recently, the UAVs as a miniature electronic products are gradually into the homes of ordinary people. Furthermore, there has been vigorous development in industrial applications, such as modern agricultural plant protection, power inspection, logistics transportation and so on [1-2]. Obviously, UAVs have a bright vision and deserve to be researched. In order to satisfy the needs of all walks of use and improve efficiency, multi-UAV cooperative flight is a topic, which can not be avoided. Compared to a single UAV, multi-UAV cooperative performing tasks will have the following advantages : (1) improving the quality of the task through mutual cooperation; (2) the ability to extend tasks by sharing resources and information; (3) shortening the task completion time and improving the efficiency of task execution , through the parallel execution of tasks; and (4) increasing the probability of successful completion by task reallocation, when platform damage occurs.

UAVs cooperative flight is an application of multi-agent cooperative formation control. After decades of development, basic cooperative control research is quite mature, and multi-UAV formation control method has achieved remarkable results. Currently the most mature approach is the Leader Follower method, i.e., an aircraft as a leader flies according to a fixed flight path, and other aircraft as followers follow the leader by taking relative position information to the leader as reference. The method was first proposed in [3] about formation control law research of the mobile robot. Although it simplifies the control strategy for multi-UAV system, but there are still

some shortcomings, such as when the leader is out of order, multi-UAV system is unable to maintain normal formation, and if there is a larger disturbance, the errors will accumulate. For this reason, many improved methods have been proposed, such as the virtual leader strategy proposed by West Virginia University. According to this algorithm, each aircraft receives the track of the virtual leader. When the fleet moved, UAVs just need track the movement of the virtual pointing to maintain the formation [4-6]. However, the virtual leader strategy lacks the distance information, which leads the difficulty of collision avoidance and large amount of communication is required, thus fault tolerance and reliability of this strategy are unsatisfactory. Literature [7] proposed a multi-UAV collision avoidance path planning method based on rolling horizon strategy, each UAV in the process of considering path coordination history and other UAV network in a moment, once found the possibility of collision, UAV immediately adjust the flight direction of collision avoidance. Fasano G et al designed two avoidance strategies to achieve autonomous collision avoidance between two UAVs by using multi-sensor fusion information, but it has higher requirements for real-time information processing ability of the system [8].

In this paper, the basic strategy of the Leader Follower is improved, the segmented waypoints GPS information and distance information based on the UWB (Ultra Wide Band) ranging module are combined to achieve the multi-UAVs cooperative flight. The cooperative algorithm consists of two main parts: the longitudinal distance information obtained from UWB is processed by the ESO (extended state observer) for reducing the disturbance and collision avoidance; the transverse correction algorithm guarantees the consistency of the UAV and prevents the follower from deviating from the track of the leader.

Compared to the original Leader Follower strategy, the complex communication protocols and higher real-time information processing ability are unnecessary. Besides, it can achieve cooperative flying even when there exists a larger disturbance or some load. The simulation and actual results show that this cooperative algorithm can achieve the two UAVs cooperative flight and complete short distance material transportation.

2 Method

2.1 The relative distance control based on ESO

By using the ESO, the disturbance generated during the flight, such as gusts, load change can be observed. In this way, the system can show strong adaptability and robustness when the parameters of the controlled object change greatly or there is a large external disturbance. In view of the influence of the gust and other disturbances during the flight, the distance acquired by the UWB can not be effectively utilized, especially when the cooperative handling load exists. Simply applying PID control law is difficult to meet the control requirements of the system. The object controlled is the relative position between the UAVs, and the observed object is the distance between the aircraft, assuming that the distance d on the first-order controlled object is:

$$\dot{d} = f(d, w, t) + b_0 u; \quad (1)$$

Wherein, u is the control input, d is the output from the system, f is the unknown function related to the current distance and disturbance, it is the total disturbance of the system, which includes the coupling effect of the system and other systems, uncertainties or variable parameters of the system itself, external disturbance and so on.

ESO is the core of ADRC control theory proposed by Han Jingqing [9], which is according the system output d and the system control input u to achieve real-time observation of system output and disturbances. Such observables $z_1 \rightarrow d$, $z_2 \rightarrow \dot{d}$ (expanded state observations), so $u = u_0 - z_2 / b_0$, substitute them into the original system [10]:

$$\dot{d} = b_0 u_0 \quad (2)$$

ESO complete formulas are as follows:

$$e = z_1 - d \quad (3)$$

$$\dot{z}_1 = z_2 + b_0 u - \beta_1 fal(e, a_{11}, \delta_{11}) \quad (4)$$

$$\dot{z}_2 = -\beta_2 fal(e, a_{22}, \delta_{22}) \quad (5)$$

$$\varepsilon = dist - z_1, \varepsilon_1 = \dot{z}_1 \quad (6)$$

$$u_0 = k_p \varepsilon + k_d \varepsilon_1 \quad (7)$$

$$u = u_0 - \frac{1}{b_0} z_2 \quad (8)$$

β_1 and β_2 are ESO gain, k_p and k_d is the gain of the PD control, b_0 is the disturbance compensation factor. $dist$ is the desired distance between the UAVs. Power nonlinear function form $fal(u, a, h)$ are:

$$fal(u, a, h) = \begin{cases} \frac{u}{h^{1-a}} & |u| \leq h \\ |u|^a sign(u) & |u| > h \end{cases} \quad (9)$$

Wherein, a is typically selected constant between 0-1, if a is smaller, the tracking process will be faster, but the filtering effect will deteriorate; δ is a constant that affects the filtering effect. The larger the δ , the better the filtering effect, but the tracking delay increases.

2.2 The transverse correction algorithm

In cooperative flight, due to the route deflection and external disturbances, the relative distance information provided can only maintain the distance between the aircraft and can not ensure that followers follow the route of the leader. It is necessary to correct the distance of its transverse offset course.

As in Figure2, suppose the starting point and target position are S and D respectively. The corresponding latitude and longitude are (LA_0, LO_0) and (LA_n, LO_n) . The route is divided into $n+1$ intermediate waypoint. The latitude and longitude of P_i th waypoint is (LA_i, LO_i) . In addition, the current position of the aircraft is $P(LA, LO)$. The transverse correction algorithm is that within each segment in the flight route, according to the current position information and previously recorded waypoint information, calculate the distance of the offset trajectory for correction. The specific algorithm is as follows:

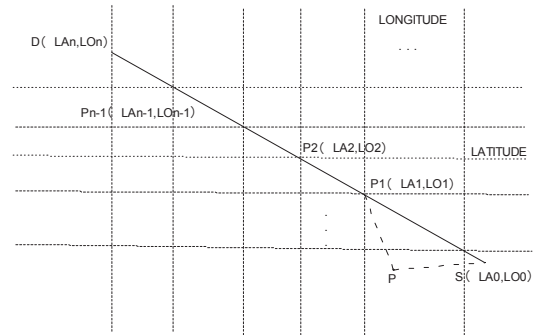


Fig 1. The transverse correction algorithm

$$\cos \varphi = \frac{\overline{PP_{i+1}}^2 + \overline{PP_i}^2 - \overline{PP_{i+1}}^2}{2\overline{PP_{i+1}}\overline{PP_i}} \quad (10)$$

$$\Delta = \overline{PP_i} \sqrt{1 - \cos^2 \varphi} \quad (11)$$

$$u_{com} = \lambda \Delta \quad (12)$$

$$\text{sgn}(\lambda) = \text{sgn}(\dot{\Delta}) \quad (13)$$

According to the segmented waypoints information, the offset angle φ can be computed. Δ is the calculated transverse offset and λ is the correction coefficient. λ is a

parameter related to the $\dot{\Delta}$ and its rule of signs is consistent with $\dot{\Delta}$. This is solved because the offset Δ is the absolute distance, if these drifts from one side to another side, the speed of the reverse problem occurs. It is determined based on the rate of change of Δ . Thus, the transverse correction amount u_{com} can be obtained through the calculated transverse offset.

2.3 Cooperative control algorithm

During the flight of multiple UAVs, the core of the cooperative control algorithm is how to guarantee the consistency in their flight, such as the two aircraft need to maintain a consistent speed and a fixed distance. By using the local distance information obtained by UWB and the global information obtained by GPS, the cooperative flight algorithm can realize the cooperative flight control of two aircraft and carry out a simple cooperative handling task. The multi-UAV cooperative flight algorithm is as follows:

$$v_1 = \bar{v} \quad (14)$$

$$v_2 = R_b^e v_2' \quad (15)$$

$$v_2' = -u' \quad (16)$$

$$u' = u + u_{com} \quad (17)$$

Wherein, v_1 and v_2 are the speed of the first and second aircraft respectively. u' is the final control output, the control output comprises a longitudinally relative distance measurement u and the compensation amounts u_{com} based on transverse correction information. Considering the existence of route deflection error in cooperative flight process, the velocity of the body coordinate is decomposed into the x and y axes of the world coordinate system according to the coordinate transformation matrix R_b^e . This ensures that the two aircraft can be stably coordinated along the established route.

In fact, the speed of the follower UAV can be decomposed into the speed along the leader route and the speed perpendicular to the route. The control algorithm provides the speed information for cooperative flight, and the actuators control the follower based on this information. This algorithm could be extended to more than two UAVs. The general control algorithm form is shown as below:

$$v_L = \bar{v} \quad (18)$$

$$v_i = R_b^e v_i' \quad (19)$$

$$v_i' = -u_i' \quad (20)$$

$$u_i' = u_i + u_{icom} \quad (21)$$

v_L is the speed of the leader, the speed of the i th UAV v_i is determined by the speed along the leader route u_i and the compensate speed u_{icom} according the neighbor UAVs' relative distances in that situation. Because of the limitation

of the experimental conditions, only two UAVs are used to verify the algorithm.

In summary, we can draw a complete multi-UAV cooperative flight control algorithm. According to the information of the relative distance measured by UWB combined with GPS information, the segmented rectification ensures that the leading UAV and the following UAV along the established trajectory movement while maintaining the stability of the distance between the aircraft to prevent collision and other hazards.

2.4 UWB distance measurement principle

UWB wireless communication technology is applied to high-speed data transmission in the early days. Nowadays, most researchers have been focusing on the UWB indoor positioning applications [11], UWB is applied to get the distance between the UAV units in this paper. It can measure the distance information timely and accurately, meanwhile ensure the communication range. UWB is known as a non-carrier communication technology, it can use nanosecond or even picosecond of non-sine wave narrow pulse for data transmission. A signal can be transmitted at low power over a wide frequency spectrum through it, the data transmission speed can up to hundreds of Mbit / s or even to several Gbit / s around 10m distance range. It has the advantages of strong anti-jamming performance, high transmission rate, large system capacity and small transmission power. In recent years, it has been widely used for short-range indoor precise positioning. The distance information obtained from UWB is taken by the TOF ranging method. The TOF ranging method is a bidirectional ranging technique that measures the distance between nodes using the propagation time of the signal between two asynchronous transceivers. Because in the line of sight environment, the distance measurement based TOF is a linear relationship with the distance, the result will be more accurate.

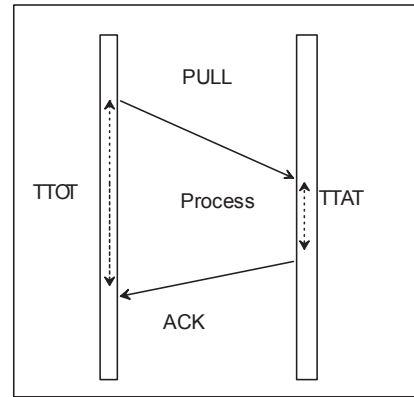


Fig 2. The TOF ranging method

The time interval between sending packets from the sending end and receiving responses is recorded as T_{TOT} , and the time interval of receiver receiving packets and sending response is recorded as T_{TAT} , as in Figure 1. Then the data packet in the air one-way flight time T_{TOF} can be calculated as:

$$T_{TOF} = \frac{T_{TOT} - T_{TAT}}{2} \quad (22)$$

Then according to T_{TOF} and electromagnetic wave propagation speed c , the distance between the two points can be calculated as:

$$d = c \times T_{TOF} \quad (23)$$

3 Simulation and results

ESO based distance control strategy, compared with the basic PID control; the simulation results show that it is more robust to some degree, as Figure 3 shows. By applying the ESO, the UAVs system can get a better performance when there is a dust or a load change.

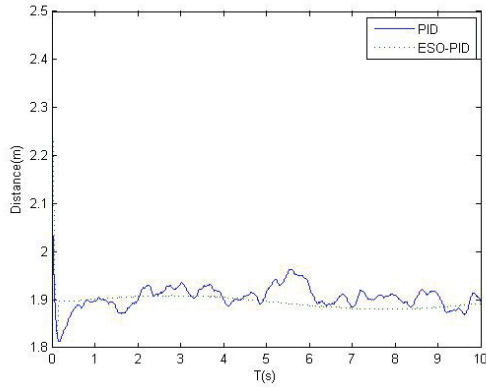


Fig 3. ESO observations simulation

According to the segmented GPS waypoint information to correct the position of the UAV, that is, from time to time on the current GPS position of the UAV, through the combination of UWB measured distance and fixed waypoints information to update the current position while not deviating from the leader's track. The two UAVs' GPS trajectory information is shown as the following figures:

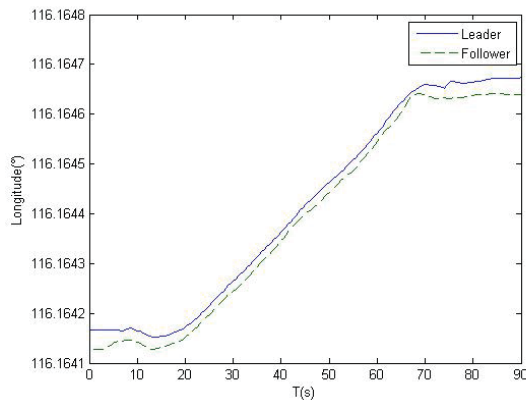


Fig 4. The longitude of the leading and following UAV

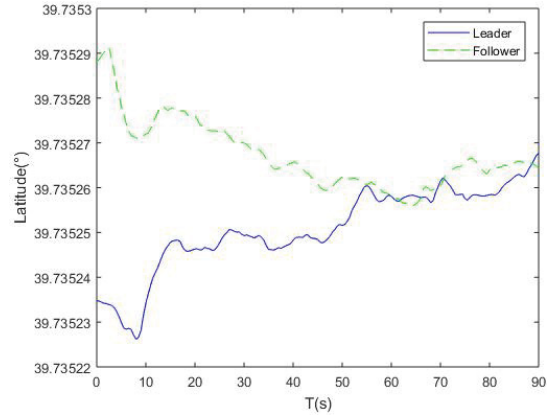


Fig 5. The latitude of the leading and following UAV

Due to the UAVs flew from west to east, the longitude of the two UAVs are almost synchronous, which means the longitudinal distance between the UAVs is stable, and the longitudinal distance is around 3.5m. The latitude of the two UAVs can reflect the transverse distance of them to some degree. The actual distance can be calculated according to the GPS information. As in Figure 6, the transverse distance of the UAVs is larger at first. It can be known that the transverse distance is limited into a narrow range gradually around 1m during the flying test, although the drift of GPS causes transverse position vibration.

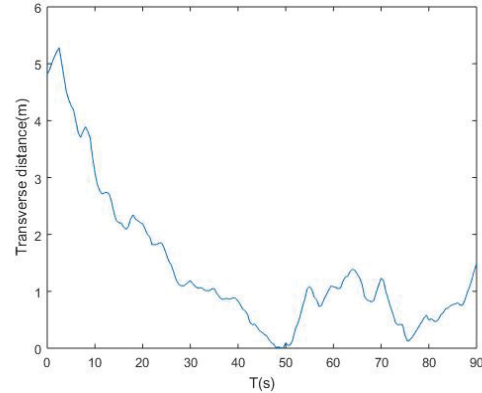


Fig 6. The transverse distance

In the case of a specific experiment, two UAVs were used, one of which was navigated at a fixed speed according to the segmented GPS waypoints, while the other aircraft followed a previous flight at a fixed distance according to this cooperative flight algorithm. After several experiments, the two aircraft can be a good cooperative flight. Further, in order to test the anti-disturbance ability and application of the cooperative flight, we also tested the suspension load (about 3 kg), and the experimental results were satisfactory, as Figure 7 shown.



Fig 7. Actual cooperative carrying task

4 Conclusions

This improved cooperative flight algorithm based on Leader Follower strategy only uses the relative distance between the UAVs and the waypoints GPS information. Compared by using complex inter-machine communication protocol to transmit desired posture, position, motion state and other information, in practice, it is easy to implement. Through the simulation test and related experiments, the cooperative flight of two UAVs can be realized, and the UAVs even could carry certain items cooperatively. The cooperative flight algorithm proposed in this paper can be extended to cooperative flight of two or more aircraft in the future to realize the cooperative task such as formation and aggregation. If the UWB's precise positioning technology is combined, it can provide higher flight precision and make up the GPS positioning information shifts.

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